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Microplastics influence on Hg methylation in diverse paddy soils

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ABSTRACT

Microplastics are widespread in estuarine, coastal, and deep sea sediments. The influence of microplastics on mercury (Hg) methylation in paddy soils with different characteristics, however, has not been well reported. In this research, we conducted a microcosmic experiment using red soil and alkaline soil with 2%, 7% and 10% polyvinyl chloride microplastics (PVC-MPs). Diffusive gradients in thin film (DGT) were used to test bioavailable Hg^{2+} and bioavailable methylmercury (MeHg) in soils. Results showed that PVC-MPs could decrease bioavailable MeHg concentrations both in red soil and alkaline soil. We demonstrated that these decreases could be due to three possible mechanisms: (1) PVC-MPs affected DOM composition, which resulted in a difference in combining capacity for bioavailable Hg^{2+}; (2) PVC-MPs decreased MeHg via changing soil properties (including sulfate and dissolved Fe); (3) PVC-MPs affected the abundance of *Proteobacteria, Firmicutes,* and *hgcA* gene in soils. Our results emphasized the significance of investigating effects of microplastics on specific contaminants to implement effective environmental remediation strategies in polluted paddy soils.

1. Introduction

Human health threat from methylmercury (MeHg) exposure is currently a global concern (Driscoll et al., 2013). Compared to fish MeHg exposure, dietary consumption of rice is a more important exposure route for humans with many populations dependent on rice for staple food (Du et al., 2018). A recent research proved children who live near the Hg mining relying on rice as staple food, with hair Hg $\geq$1 μg g^{-1} were 1.58 times more likely to have an IQ score <80 (Feng et al., 2020). Consequently, factors that may influence Hg methylation in paddy soils would then likely impact on human exposure to Hg.

Microplastics (plastic particles <5 mm) in the environment, especially in agricultural systems (Boots et al., 2019a; Blaesing and Amelung, 2018) (sources include atmospheric deposition, irrigation with wastewater, sewage sludge addition, compost or fertilization, and the use of mulching film) have attracted a lot of attention (Rillig, 2020). Plastics production increased from 5 Mt a^{-1} to 359 Mt a^{-1} from 1950s to 2018, and the cumulative production will reach 34,000 Mt by 2050 (Thompson et al., 2009; Zhang et al., 2021). Notably, the low recycle rate makes 20–42% of plastic waste remained and long-termly existed in soils. It has been confirmed that up to 380 – 3786 pieces kg^{-1} microplastics were found in agricultural soils (Li et al., 2021). This large input of microplastics in soils has enhanced global research interest in investigation the impacts of microplastics on soil biophysical factors such as water holding capacity, soil bulk density (Machado et al., 2018; Yang et al., 2021), dissolved organic matter (DOM) composition (Liu et al., 2017), microbial communities (Seeley et al., 2020), plant growth (Boots et al., 2019b), as well as metal transformation/bioavailability, while potential impacts of microplastics on Hg biogeochemistry are largely ignored. Considering that they can leach DOM and serve as organic carbon substrates for soil microbial communities, microplastics have potential to influence the bioavailability of Hg and production of net MeHg in soils. However, the impact mechanism of microplastics on Hg in paddy soils, especially different effects of diverse paddy soils, has yet to be studied.

Generally, the amount of inorganic Hg (Hg^{2+}) which is availability to methylation microorganisms is one of the most key factors for regulating Hg^{2+} methylation rates in anaerobic settings (Hsu-Kim et al., 2013a). The diffusive gradients in thin film (DGT), a widely published technique, showed better performance in assessing bioavailable Hg, compared to conventional filtration and thiol-based extraction approach (Rivera

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et al., 2019; Ndu et al., 2018). Many reports have documented successful applications of DGT for probing bioavailable Hg^{2+} and MeHg in water (Fernandez-Gomez et al., 2015), sediment (Hong et al., 2014) and soil (Turull et al., 2019) in both field and laboratory tests. In our research, we investigated the influence of microplastics on Hg methylation by using DGT to quantify bioavailable Hg^{2+} and MeHg in soils, and comparing the reaction mechanisms in diverse paddy soils. We hypothesized that microplastics affect Hg methylation due to 1) soil properties (including DOM composition and redox condition) and 2) methylation microorganisms. To test our hypothesis, polyvinyl chloride microplastics (PVC-MPs, a high-density polymer) with different concentrations were added to both red soil and alkaline soil (representative arable soils commonly found in China) and incubated for 21 days, to evaluate the short-term changes of Hg methylation in paddy soils. Fluorescence EEMs and PARAFAC were measured to infer the influence of PVC-MPs on quantities and composition of paddy soil DOM. MiSeq sequencing of 16S rRNA genes and quantitative polymerase chain reaction (qPCR) were used to assess the changes in diversity and composition of microbial communities and methylation function genes in two paddy soils.

2. Material and methods

Samples used in this study were collected from agricultural fields. Red soil was obtained from Yongzhou, Hunan Province. Alkaline soil was from Chifeng, Inner Mongolia. They were classified according to Chinese soil classification (Yan et al., 2020). The soils at a depth of 0–10 cm were used in these experiments, which were air-dried, mixed, sieved to diameter ≤2 mm, and stored in greenhouse (~22 ℃) before experiment. The basic characteristics of soils are shown in Table S1.

PVC-MPs powder of 0.1–10 μm diameter was acquired from Lingfeng company, Wuxi, China. Characteristics of microplastics are shown in Table S2. Microcosmic examination committed these microplastics were reasonably spherical and had diameters with little deviation from nominal size. The choice of PVC-MPs used in this study was based on a previous report that PVC beads were likely to effectively permeate into soil matrix (Machado et al., 2018). 2%, 7%, 10% of PVC-MPs were added at soil dry weight in microplastics amendment and a control treatment group with no microplastics addition (nominal concentration 0%) (Fig. 1). Lower limit concentration of microplastics (2%) was decided grounded on no visual distinction could be detected in paddy soil after microplastics were added, that is, these particles entirely blended into soil matrix (Machado et al., 2018). 7% concentration was the reported level of pollution associated with the environment (Machado et al., 2018; Fuller and Gautam, 2016), 10% concentration was intentionally designed to be higher than the values reported in natural soil environment (Alimi et al., 2018). Although no test microplastics were added in control treatment, it did not represent the absolute absence of microplastics in soil. In fact, since the soil was originally collected from surrounding areas of the city, the actual content of microplastics in control sample may be close to or slightly lower than minimum concentration of microplastics added.

2.1. Experimental set up and design

For each experimental replicate, 50 g soil or soil-microplastics mixture was added to glass jars of 6.2 cm height and 6.9 cm diameter, with 100 ml deionized water (18.2 MΩ cm Milli-Q water). Each sample was amended with 1 mM sodium pyruvate to stimulate methylation microbial community (Ndu et al., 2018). Glass jars were sealed with gastight screw caps, wrapped in tin foil, and incubated at room temperature in dark for 4 days. An additional control treatment with a resazurin indicator (0.001%) altered from dark pink to colorless during this time, indicating the anoxic conditions in jars (Ndu et al., 2018). After this preincubation period, all further manipulations of the experiments were performed in an anaerobic operating chamber.

Target Hg content was 800 ng g^{-1} d.w. Jars were stirred for a few seconds before Hg was added, a DGT sampler was then placed in each glass jar stored under room temperature (25 ℃). Introduction and calculation of DGT are described in Supporting Information. In order to avoid photodegradation of MeHg, all jars were cultured in dark environment. Replicate jars (n = 3) were sacrificed for sampling at designated contact times <1 and up to 21 days after Hg addition. After sample collection, a large amount of deionized water was applied to rinse the particles adhered to DGT filter, which could prevent further accumulation after retrieval. A Ziploc bag with 2 ml deionized water was used to prevent moisture evaporation and oxidation of DGT probe before analysis (Wang et al., 2017).

2.2. Index analyses

Oxidation reduction potential (ORP) and pH (by HQ30d multiparameter meters, HACH, USA) were tested at designated incubation time point, and then each jar was stirred to mix and held static for 10 min to

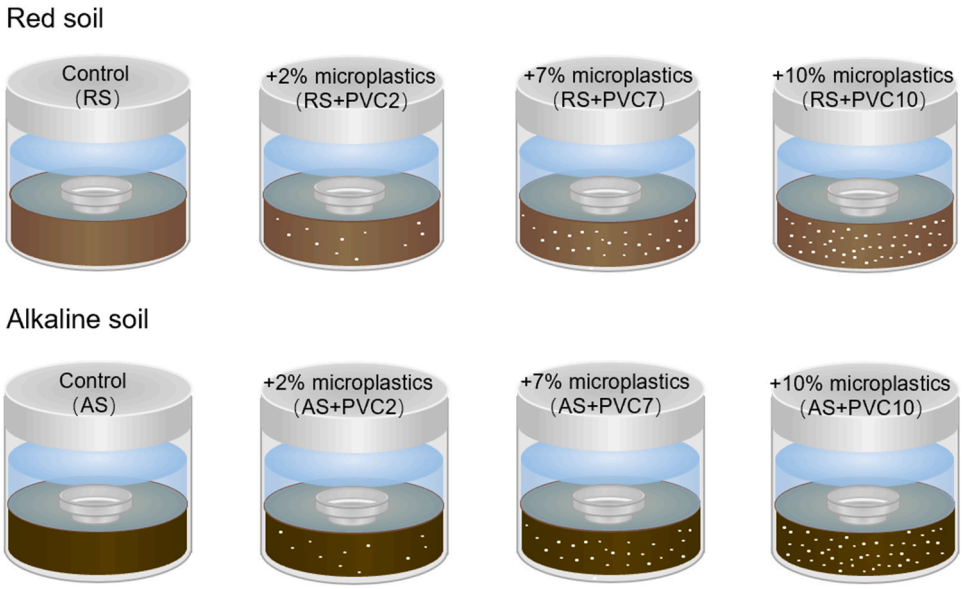


Fig. 1. Microcosmic experimental design, all eight treatments were repeated in triplicate.

allow for large particles settle prior to subsampling in anaerobic chamber. Overlying water was collected by filtering 30 ml through a 0.2 μm poly(ether sulfone) filter. This filtered sample was apportioned into samples designated for major cations analysis including dissolved Fe (Aliquots of filtered overlying water were acidified with 15.8 M trace metal grade nitric acid and then analyzed for dissolved Fe by ICP-MS), and other aliquots were analyzed for sulfate concentration by ion chromatography (Dionex ICS2100 ion chromatograph with AS18 column).

DOM levels in water samples was represented as dissolved organic carbon (DOC) in mg L^{-1}, which were analysed using standard soil test procedures of Chinese Ecosystem Research Network (CERN Editorial Committee, 1996). DOC concentrations were determined using a TOC analyzer based on high-temperature catalytic oxidation (Shimadzu L-series, TOC-CHP, Japan). Excitation (Ex) and emission (Em) scanning ranges of excitation emission matrix (EEMs) were set to 200–450 nm (10 nm interval) and 250–502 nm (4 nm interval), respectively. Rayleigh scatter peaks were eliminated by cut and subsequent linear interpolation using drEEM toolbox (Murphy et al., 2013). Parallel factor analysis modeling (PARAFAC) was performed using MATLAB 2019a with drEEM Toolbox. The number of independent fluorescent DOM components was determined using split-half validation and percentage of the explained variance (99.6%). Relativity fluorescence intensities (Fmax) (A.U.) of identified FDOM components were used to represent relative proportions of the components.

DGT from each jar was disassembled at sampling time point, stable elution liquid were obtained for MeHg and total Hg using a mild reagent containing 0.1 M HCl and 2% thiourea (Ren et al., 2018). Dilutions of these extracts were measured using cold vapor atomic fluorescence spectrometry (CV-AFS). (The difference between the total Hg and MeHg, was used to yield Hg^{2+}).

To explore the changes of soil microbial communities and methylation genes during culture, paddy soils collected at 1, 3, 7, 11 and 21 days were chosen for high-throughput 16S rRNA sequencing and real-time PCR detection (Supporting Information).

2.3. Statistical analysis

The structural equation modeling (SEM) was constructed using Smartpls 3.0 to evaluate direct and indirect effects of bioavailable Hg^{2+} and MeHg, soil properties (pH, ORP, DOC, sulfate, dissolved Fe), DOM composition and methylation microorganisms (community, diversity and *hgc*A copy number). SEM is particularly useful in investigating complex relationships among predictors commonly found in environmental ecosystems, as it allows the partitioning of causal influences among multiple variables and the separation of direct and indirect effects of model predictors (Grace, 2006). The aim of SEM used in this research was to identify decisive difference of soil properties and microbial communities attributed to Hg methylation after microplastics addition in two diverse paddy soils. Before model calculation, we established a priori model according to our current knowledge of the impacts of environmental variables on bioavailable Hg^{2+} and MeHg (Fig. S1).

We used maximum likelihood evaluation procedure to compare the constructed SEM model with the observation. The model has a good fit with the observation judged through a non-significant χ^2 test ($P > 0.05$), akaike information criterion (AIC < 0.05), goodness-of-fit (GFI > 0.90), and root square mean error of approximation (RMSEA < 0.05).

3. Results

3.1. Bioavailable Hg^{2+} and MeHg content on DGT

The concentrations of bioavailable Hg^{2+} and MeHg were measured using DGT in each sampling bottle (Figs. 2 and 3). The soil bioavailable Hg^{2+} and MeHg decreased over time in red soil without PVC-MPs(RS). After adding PVC-MPs, bioavailable Hg^{2+} showed an upward trend from 11 days ($p = 0.003$), while bioavailable MeHg still decreased with increasing cultivation time ($p = 0.024$); The soil bioavailable Hg^{2+} and MeHg increased over time in alkaline soil without PVC-MPs(AS). After adding PVC-MPs, bioavailable Hg^{2+} and MeHg remained relatively stable over 21 days experiments.

In red soil, bioavailable Hg^{2+} and MeHg in RS showed opposite trends. When PVC-MPs were added as 2%, 7% and 10%, bioavailable Hg^{2+} were 55.20%, 37.37% and 61.98% higher than that in RS after 21 days, respectively, while bioavailable MeHg were 8.49%, 8.36%, and 21.67% lower than that in RS after 21 days, respectively. In alkaline soils, bioavailable Hg^{2+} and MeHg in AS showed the same trend, when PVC-MPs were added as 2%, 7% and 10%, bioavailable Hg^{2+} was 67.70%, 51.36%, and 48.53% lower than that in AS after 21 days, respectively; Bioavailable MeHg was 35.02%, 58.42%, and 43.40% lower than that in AS after 21 days, respectively. These results suggested that the influence of PVC-MPs on soil Hg methlytion seemed to depend on soil properties.

3.2. Paddy soil characterization

Several key factors controlling Hg methylation i.e., pH, ORP, sulfate and dissolved Fe concentrations were quantified (Figs. S2–S5). Briefly, the pH of red soil increased slightly from ~5.7 to 6.8 but the pH of alkaline soil was stabilized at 7.3 without obvious change during anaerobic culture period (Fig. S2). This result can be interpreted that under flooding conditions, paddy soil which was initially acidic consumed protons due to reduction process, and soil pH gradually

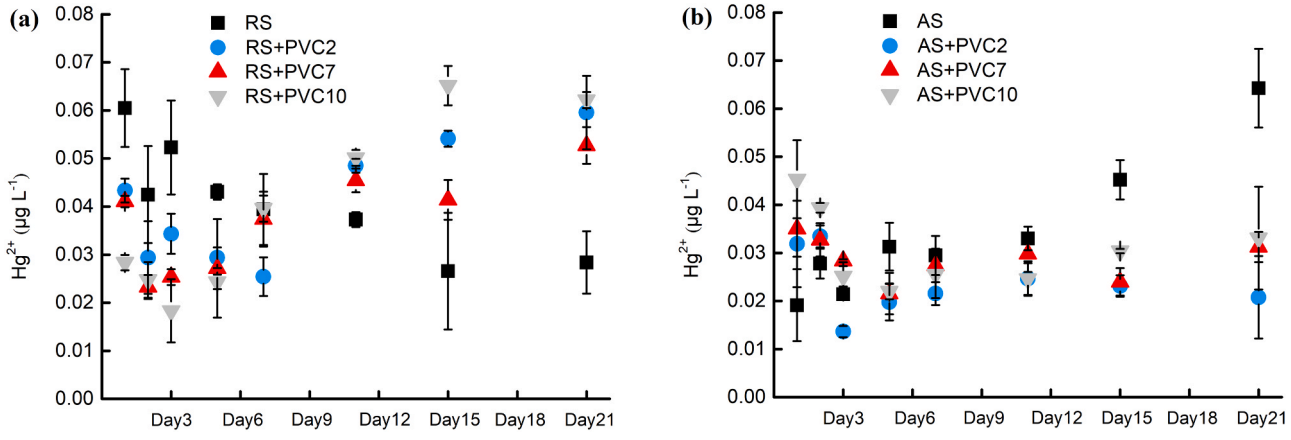


Fig. 2. Bioavailable Hg^{2+} accumulated on DGT in red soil (a) and alkaline soil (b).

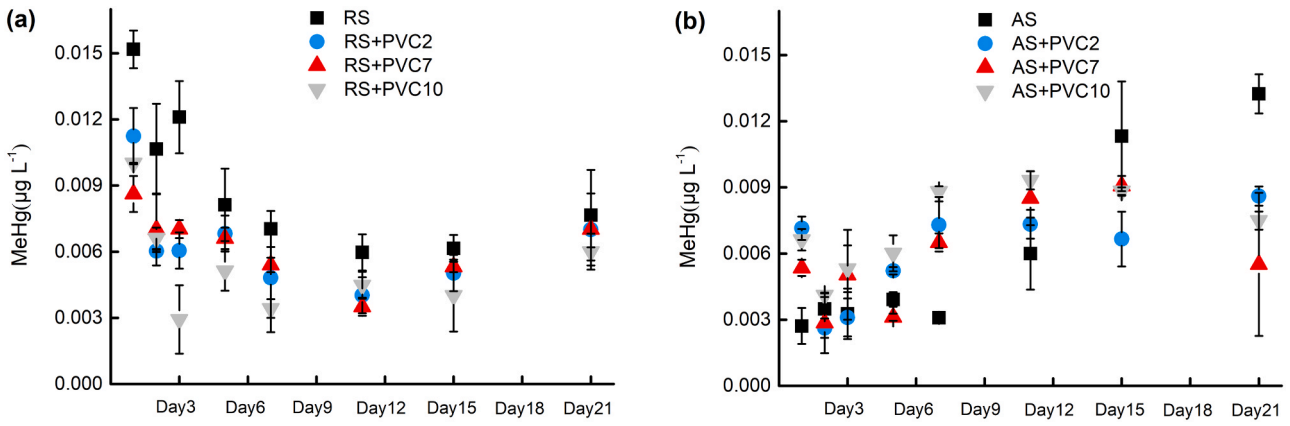


Fig. 3. Bioavailable MeHg accumulated on DGT in red soil (a) and alkaline soil (b) (Fig. S3).

increased since CO_2 causes mineralization triggers of acidification, while the contrary is true for alkaline soil (Sahrawat, 2015). ORP in red soil decreased rapidly from −100 to −300 mV during incubations, while ORP in alkaline soil always stabilized between −300 and −400 mV, indicating the effects of PVC-MPs on ORP in the system were greatly related to the content of ORP in the system.

In red soil, sulfate reduction in RS started after 2 days and completed after 7 days (from 93.8 mg L^{-1} to 32.1 mg L^{-1}, Fig. S4a), while sulfate reduction in red soil seemed to slow after adding PVC-MPs. In alkaline soil, complete sulfate consumption was observed only after 21 days (from 97.7 mg L^{-1} to 5.3 mg L^{-1}, Fig. S4b), and it was not significantly affected by PVC-MPs. Dissolved Fe in AS started increase from 11 days (0.03 mg L^{-1}) to 21 days (3.15 mg L^{-1}) (Fig. S5a), reflecting the onset of microbial reduction with Fe(III)-(oxyhydr)-oxides. At 15 days, dissolved Fe concentration of PVC-MPs added (13.51 mg L^{-1}) was twice as high as than in AS (6.02 mg L^{-1}). In contrast, dissolved Fe in alkaline soil was low at the start, and increased slightly from 7 days to 21 days (0.11 mg L^{-1} to 0.9 mg L^{-1}, Fig. S5b), while it decreased after PVC-MPs addition. These results demonstrated that PVC-MPs affected the geochemical characteristics relevant to Hg methylation.

3.3. DOM concentration and composition

3.3.1. DOC contents

The addition of PVC-MPs significantly increased the levels of DOC in the system (Fig. 4) (Seeley et al., 2020). In red soil, DOC levels increased with incubation time. At 21 days, DOC levels in RS+PVC2, RS+PVC7 and RS+PVC10 were 33.8%, 83.4% and 127.8% higher than that in RS, respectively. In alkaline soil, DOC levels in AS and AS+PVC2 did not significantly change from 1 day to 21 days, while DOC levels in AS+PVC7 (124.04 mg L^{-1}) and AS+PVC10 (179.33 mg L^{-1}) were significantly higher than that in AS (42.83 mg L^{-1}) at 2 days, but decreased consistently with time (AS+PVC7 decreased by 88.2%, AS+PVC10 decreased by 91.0%). These results indicated the microplastics may act as an organic carbon source for microbial communities in paddy soils.

3.3.2. Fluorescent EEMs of DOM

Fluorescence EEMs of DOM in red soil and alkaline soil were compared for different PVC-MPs concentrations (2%, 7%, 10% and control sample, Fig. 5). With or without PVC-MPs addition, several peaks prominently appeared in humic acid-like fluorescence region, which are typically defined as peak C (Ex/Em = 350/420–480 nm) and peak A (Ex/Em = 260/380–460 nm) (Carstea et al., 2016; Coble, 1996). Peak A may be related to hydrophobic acids substances (Murphy et al., 2011), while peak C is a common product of terrestrially derived organic matter, which is this type of component to be photorecalcitrant marker for recalcitrant DOM (Walker et al., 2009; Osburn et al., 2015). Many fluorescent compounds in paddy soils (e.g., lignins, tannins, etc.) were also observed in fluorescence EEM region with Em > 380 nm (Li et al., 2018, 2019). DOM fluorescence intensity was lower in PVC-MPs addition samples (2%: 0–1391 A.U.; 7%: 0–1645 A.U.;10%: 0–1613 A.U.) than in RS (0–2206 A.U.) in red soil, which would be the result of microplastics addition facilitating the decomposition of humic-like material (Liu et al., 2017). However, PVC-MPs added to alkaline soil were not able to produce significant changes in fluorescence intensity, while fluorescence peak shifted toward lower excitation wavelengths. These results, therefore, seemed to imply that microplastics could act as

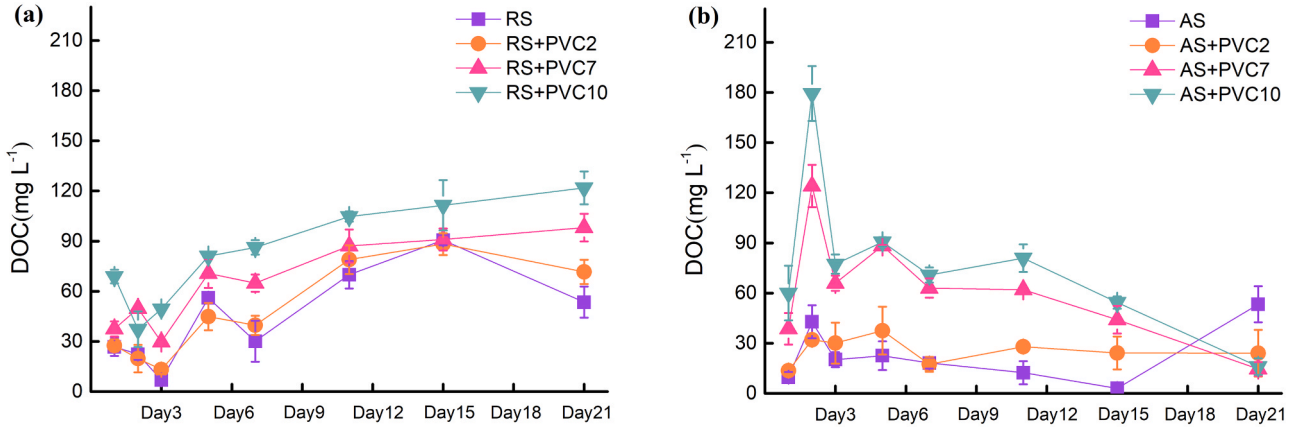


Fig. 4. Variations with time of DOC concentrations with PVC-MPs treatments.(a) red soil (b) alkaline soil

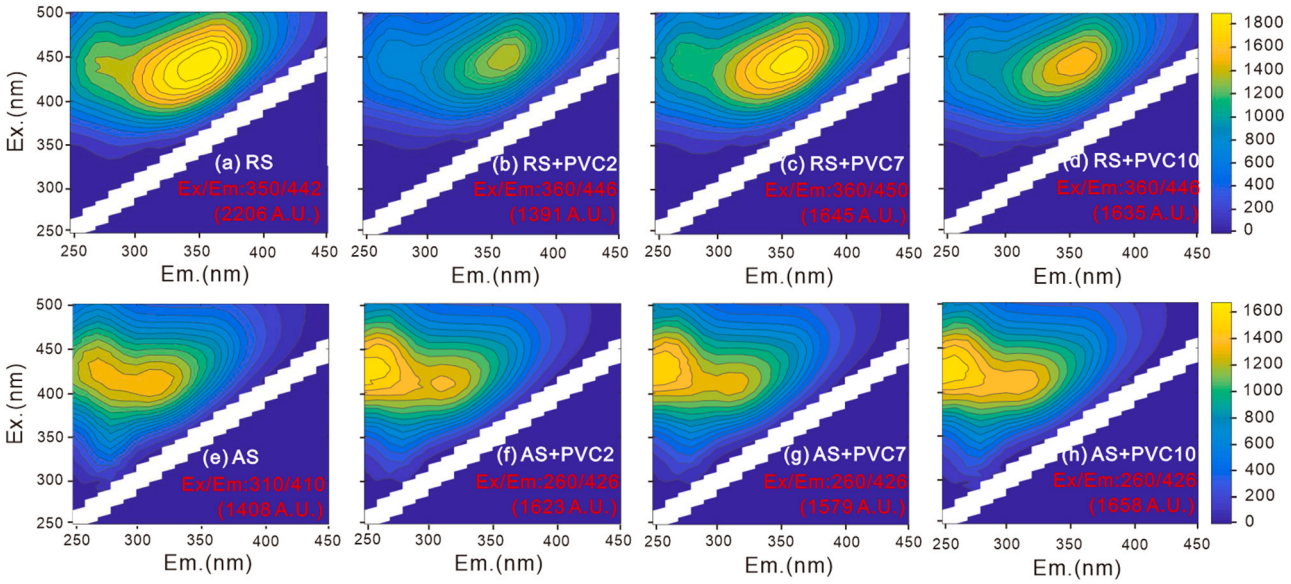


Fig. 5. Fluorescence EEM plots of DOM in red soil with RS (a), RS+PVC2 (b), RS+PVC7 (c), and RS+PVC10 (d) after 21 days; Fluorescence EEM plots of DOM in alkaline soil with AS (e), AS+PVC2 (f), AS+PVC7 (g) and AS+PVC10 (h) after 21 days.

a non-negligible source of DOM in environments after contacting with water during a period of time (Liu et al., 2017), while the fate of microplastics is greatly influenced by environmental parameters (Singh et al., 2019).

3.3.3. *Fluorescence signatures of different DOM components*

We used a four-component PARAFAC model to describe the variations of the fluorescence features in incubations (Table S3, Fig. 6). Component 1 (Comp 1) and 2 (Comp 2) proved maxima at 260(340)/426 nm (Ex/Em) and 250/434 nm, respectively. The peak locations of Comp 1, Comp 2 fell in humic-like fluorescence region, while component 3 (Comp 3) associated with fulvic acid-like substances exhibited its peaks at longer wavelengths of 275(375)/474 nm (Ex/Em), which indicating a microbially transformed substances. Ex/Em wavelength pairs of component 4 (Comp 4) were concentrated at 300/390 nm (Ex/Em), which were derived from a autochthonous protein-like substance related to microbial degradation. It is worth noting that four DOM components derived from paddy soils were partially overlapped spectrally with components derived from typical aquatic systems (seawater or freshwater).

After 21 days, the fluorescence of four components in red soil with PVC-MPs was quenched, especially the fluorescence of Comp 1, Comp 2, Comp 3, Comp 4 in RS+PVC2 decreased 45.3%, 73.0%, 24.5%, and 62.8%, respectively (Fig. 7). On the contrary, the added microplastics increased the fluorescence of humic-like Comp 2 by 31.3–36.4% in alkaline soil, in comparison to AS treatment, but other fluorescent components were not greatly affected by addition of PVC-MPs.

3.4. *Microbial analysis*

3.4.1. *Microbial 16S rRNA analysis*

Sequencing results showed an average of 55,887 sequences per sample. Bacterial α diversity increased over time with PVC-MPs added in red soil especially RS+PVC2. While in alkaline soil, both in AS and other treatments, the α diversity decreased over time (Table. S4).

The comparisons between 1 day and 21 days of bacterial community structure at phylum level in red and alkaline soils are shown in Fig. 8. There were obvious differences in community composition between the two paddy soils. Red soil was dominated by phyla *Proteobacteria, Actinobacteriota, Firmicutes* and *Planctomycetota.* Of the 4 phyla, *Proteobacteria* had the highest relative abundance in soil, accounting for 27–34% of the total number of bacteria. After 21 days incubation, *Proteobacteria* abundance in RS+PVC2, RS+PVC7 and RS+PVC10 decreased by 9.2%, 13.6% and 11.4%, respectively. At class level (Fig. S6a), relative abundance of *Alphaproteobacteria* and *Gammaproteobacteria* in soil decreased with increasing concentration of PVC-MPs, while *Planctomycetes* increased significantly with increasing concentration of PVC-MPs. Family *Isosphaeraceae* increased more significantly in PVC-MPs treatments than that in RS (Fig. S7a). Genera *Aqusphaera* increased when PVC-MPs were added compared to RS, while genera *Bacillus* decreased.

Alkaline soil was dominated by phyla *Chloroflexi*, *Proteobacteria*, *Actinobacteriota* and *Firmicutes*. These bacterial communities showed different trends with those in red soil when PVC-MPs were added, as only AS+PVC10 treatment decreased the *Proteobacteria* abundance. In AS, *Choroflexi* abundance decreased by 6.6%. The abundance of *Choroflexi* was less affected in AS+PVC2, while in AS+PVC7 and AS+PVC10, *Chloroflexi* abundance increased by 11.7% and 12.4%, respectively. Several classes showed a significantly lower relative abundance in microplastics treatments than AS community (Fig. S6b), including *Alphaproteobacteria* and *Gammaproteobacteria*, while the addition of PVC-MPs made *Planctomycetes* abundance higher than that in AS. Family *Anaerolineaceae* and *A4b* decreased significantly (Fig. S7b), while genus *norank KD4–96* increased significantly when PVC-MPs were added.

3.4.2. *Effects of microplastics on soil hgcA gene expression*

We used reliable molecular marker *hgc*A gene clusters to identify mercury methylation communities and to predict MeHg in environment to gain insights into Hg methylation condition (Podar et al., 2015). Gene copy numbers of *hgc*A were measured based on qPCR relative to 16S rRNA which was plotted in Fig. 9. Bacterial *hgc*A gene abundances ranged from 2.67×10^8 to 5.36×10^9 per gram of red soil, and 2.32×10^8 to 2.64×10^9 per gram of alkaline soil, respectively. The *hgc*A gene abundances decreased from 1 to 21 days for all treatments, suggesting reduced methylation potential with time. Bioavailable MeHg levels in red soil were corresponded with *hgc*A gene copy numbers, and RS+PVC2 group had the highest *hgc*A gene abundance. In contrast, there was no correlation between *hgc*A gene abundance and bioavailable MeHg levels in alkaline soil. HgcA gene abundance in control group was the highest at the end of incubation (21 days) in two paddy soils.

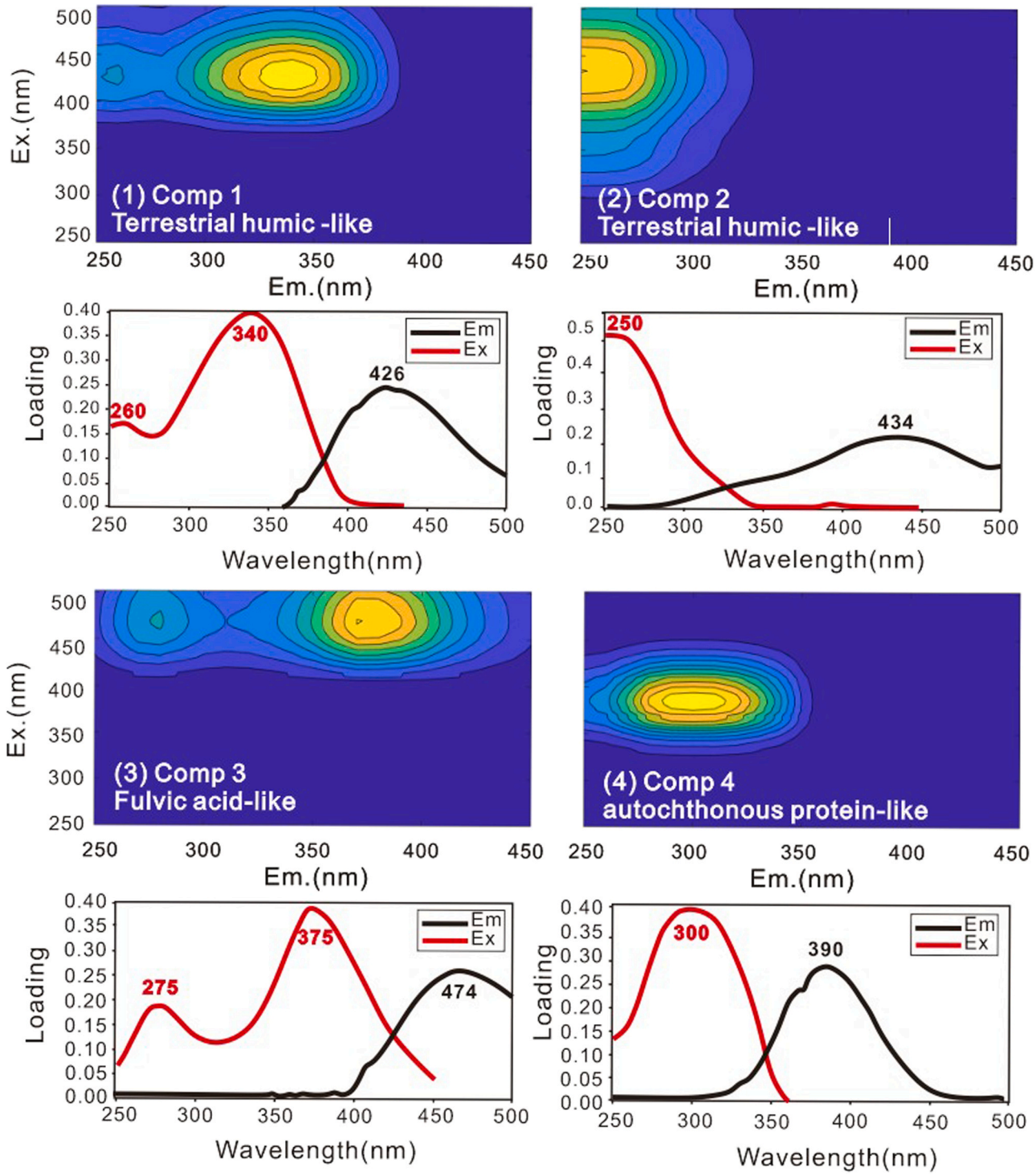


Fig. 6. PARAFAC model output showing the four different fluorescent components (Comp 1: Terrestrial humic-like, Comp 2: Terrestrial humic-like, Comp 3: Fulvic acid-like, Comp 4: Autochthonous protein-like) and corresponding excitation/emission loadings.

3.5. SEM analysis

Fig. 10(a) and (b) showed SEM analysis of red soil and alkaline soil. In red soil, SEM model explained 89.66% and 81.74% of bioavailable Hg^{2+} and MeHg by soil properties as well as bacterial community assemblage, respectively. Dissolved Fe ($r = 0.40$, $p < 0.0001$), microbial diversity and ($r = 0.25$, $p < 0.001$) comp1 ($r = 1.16$, $p < 0.05$) had direct positive effects, while sulfate concentration ($r = -0.48$, $p < 0.0001$) and comp3 ($r = -1.30$, $p < 0.05$) had direct negative effects on bioavailable Hg^{2+}. pH, DOC and comp1 had negative effects on bioavailable MeHg. In alkaline soil, SEM model explained 72.7% and 78.6% of bioavailable Hg^{2+} and MeHg, respectively. Dissolved Fe ($r = 0.88$, $p < 0.0001$), sulfate concentration ($r = 0.52$, $p < 0.0001$), microbial community ($r = 0.24$, $p < 0.0001$) and DOC ($r = 0.26$, $p < 0.0001$) had direct positive effects on bioavailable Hg^{2+}, while dissolved Fe ($r = 0.51$, $p < 0.0001$) and comp3 had direct effects on bioavailable MeHg.

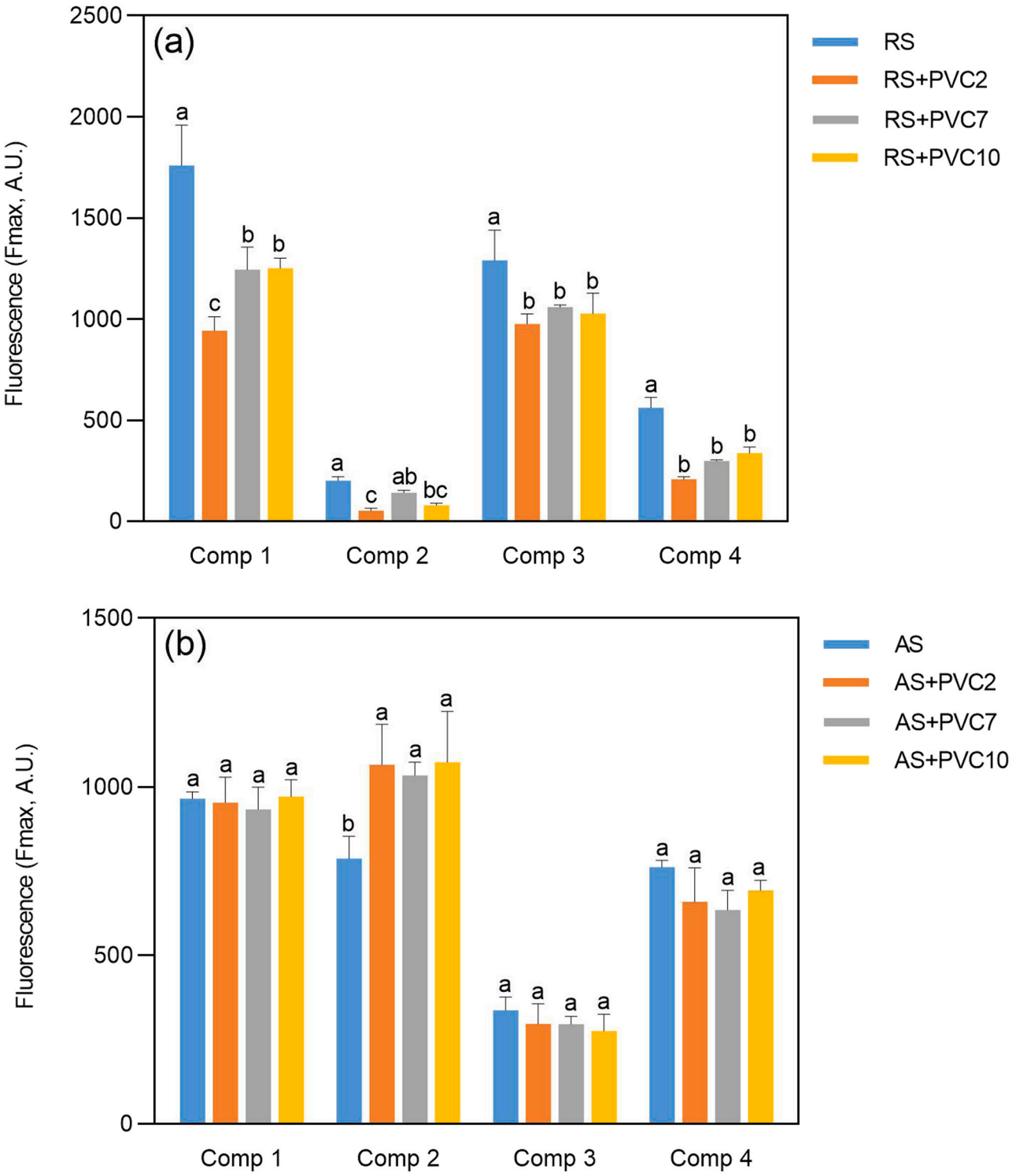


Fig. 7. Relative abundances of four fluorescent components (Comp1 to Comp4) for (a) red soil and (b) alkaline soil after 21 days.

4. Discussion

This study is the first report to demonstrate the influence mechanism of microplastics on Hg methylation in diverse flooding paddy soils. Hg extracted by DGT represents the part that can be used by methylators and other organisms in paddy soils. Our research showed that PVC-MPs contamination decreased soil bioavailable MeHg concentrations under flooding conditions (Fig. 3). More importantly, considering the less adsorption between microplastics and Hg (Singh et al., 2019; Turner and Holmes, 2015), our results proved that soil physiochemical and biological parameters decided potential impact mechanisms of microplastics on Hg methylation, with strong indirect effects on MeHg production via soil microbial attributes, rather than direct effects. Therefore, we focus on expounding and comparing the mechanisms related to biogeochemical factors.

We observed that the amount of bioavailable Hg^{2+} and MeHg decreased over time in RS treatment, whereas the result in alkaline soil was just opposite (Figs. 2 and 3), indicating that soil properties cannot

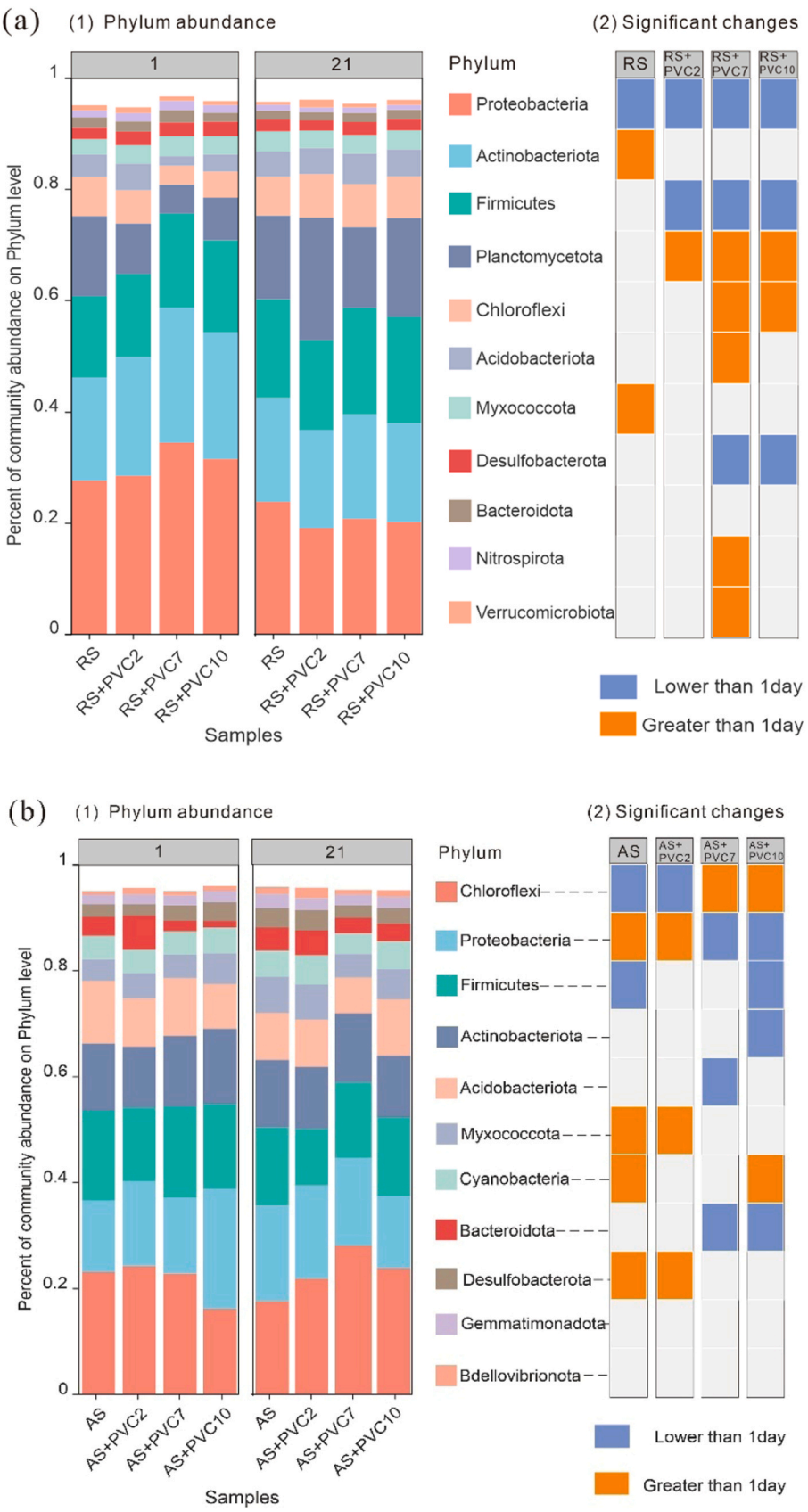


Fig. 8. Comparison of taxonomic differences (phylum level) in (a) red soil and (b) alkaline soil bacterial communities with different PVC-MPs concentration. Stacked bar plot the relative abundance of phylums (**>** 1% abundance) for each plastic treatment in each sediment collection date (1, 21 days), (1) phylums that are significantly different in relative abundance between treatments and control, determined using DeSeq, is illustrated in (2) showing if a phylum is significantly lower (blue) or greater (orange) than 1 day. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

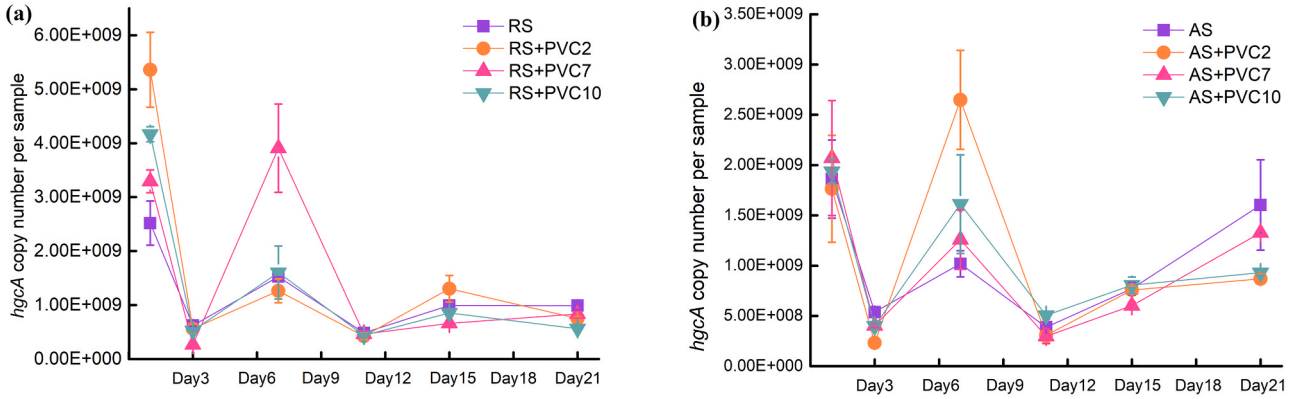


Fig. 9. HgcA abundance for *Deltaproteobacteria* in DNA extracts (0.53 g) of (a) red soil and (b) alkaline soil.

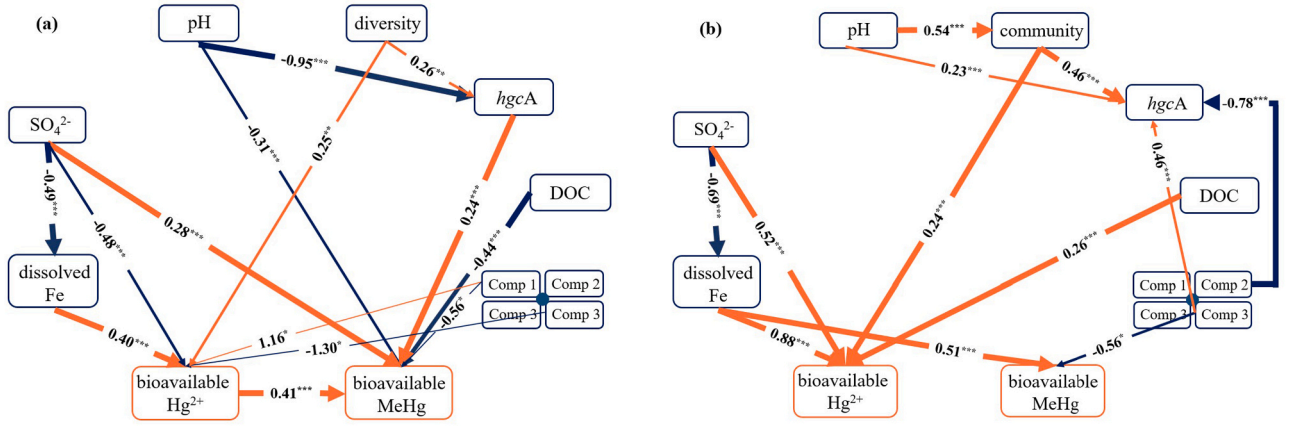


Fig. 10. Structural equation modeling (SEM) (a: red soil; b: alkaline soil) showing direct and indirect effects of soil properties and microbial community attributes on bioavailable Hg^{2+} and MeHg concentration after microplastics were added, respectively. The thickness of arrow represents the strength of the relationship, while no arrow is shown when the effect is not significant. Comp 1–4 represent four DOM fluorescent compositions, $*p < 0.05$, $**p < 0.01$, $***p < 0.001$.

be ignored in research of Hg biogeochemical cycles. Red soil was an acidic soil with relatively high ORP and low microbial diversity and abundance, while alkaline soil was with relatively low ORP and high microbial diversity and abundance. These two had obvious differences in sulfate (Fig. S4), iron reduction (Fig. S5) and microbial community composition (Fig. 8), etc., even the background values of Hg were 12 times different (Table. S1). These discrepancies were reflected in Hg contents. In fact, soil's physical and chemical properties, such as pH, ORP, organic carbon, sulfur and iron, could all influence Hg availability and methylation (Driscoll et al., 2013; Ndu et al., 2018; Wu et al., 2020a). Lui et al., (2019) observed significantly different MeHg between southwest and northern paddy soils in China. They suggested that this was attributable to the source of bioavailable Hg in soil. Moreover, differences in the composition of soil microbial community could also be the main factor. In addition, Hg methylation covaried strongly with sulfate reduction in time and space, and blocking sulfate reduction typically led to dramatically lower MeHg production (Ndu et al., 2018). Our SEMs showed that sulfate reduction had negative and positive effects on bioavailable Hg^{2+} in red and alkaline soil, respectively (Fig. 10), while dissolved Fe had a strong positive effect on bioavailable Hg^{2+}. These two may be part of the reasons for the difference in bioavailable Hg^{2+} changes between the two soils.

Based on both fluorescence EEMs and PARAFAC analysis (Figs. 5 and 6), PVC-MPs treatments led to significant changes in DOM composition of two paddy soils. As can be seen from the results, four DOM fluorescence components were markedly decreased or increased in red soil and alkaline soil, reflecting the difference in role of microplastics. As well as correspondingly, DOC levels in both two soils showed increasing trends with the dose of PVC-MPs increasing (Fig. 4), which affected carbon cycle in our microcosms. However, DOC levels in AS+PVC7 and AS+PVC10 decreased over time. We considered the reason that microbes would have efficiently taken up the microplastic-leached DOC immediately upon its release. Considering that microplastics are carbon-based chain structures, they could act as a new source of organic carbon and leach detectable amounts of DOC (both polymers and additives) into aquatic ecosystems, and then influence the existing carbon cycle (Lee et al., 2020). Even the bioavailable fraction of the leached DOC was higher when the plastic was kept in dark than solar radiation (Romera-Castillo et al., 2018). Romera-Castillo et al. (2018) demonstrated that dissolved organic carbon pool and greenhouse gas production (methane, ethylene, etc.) (Royer et al., 2018) in marine waters were affected by chemical substances extracted from plastic, so that microplastics may become one of the organic carbon sources of sediment microbial communities. Another possible reason was microplastics could also contribute to DOC measurement directly through their presence in the samples. DOC is operationally defined as the fraction of organic matter passing through a 0.2–0.7 μm filter. Therefore, any microplastics smaller than 0.7 μm would pass the most commonly used filters for DOC analysis and thus contribute to measured DOC (Romera-Castillo et al., 2018). In most settings water stable Hg concentration (bioavailability) was observed to increase with organic carbon content in soils, which was found to be the fundamental factor via its complexation with Hg and stimulating effect of the labile carbon on microbial growth (Hu et al., 2019; Hsu-Kim et al., 2013b). The same result of this research was in red soil, in which DOC had a positive effect on bioavailable Hg^{2+}. In alkaline soil, DOC had a negative effect on bioavailable MeHg. This may be

caused by the formation of DOC-MeHg complexes with low bioavailability. In this case, DOM increased the stability of MeHg in treatments (Taylor et al., 2019).

Insights into the mechanisms behind microplastics impacts on soil and aquatic DOM drawn from other studies. Lee et al. (2020) established that a prolonged contact of commercial plastics with water may cause the dissolution of the additives and the oligomers/monomers of the plastics (Suhrhoff and Scholz-Boettcher, 2016), which cause enhanced humic-like fluorescence. Indeed, our PVC-MPs treatment in alkaline soil also increased the Comp 2 after 21 days (Fig. 7). However, considering the dual effects of microplastics on water and soil in flooding paddy soils, it cannot be ruled out stimulating the activity of enzyme, which results in the enhancement of microbial hydrolytic activity DOM, and then leads to DOM accumulation (Liu et al., 2017). In contrast, the humic-like, fulvic acid-like and autochthonous protein-like peaks appeared to be inhibited with PVC-MPs addition in red soil. Drosos and Piccolo (2018) showed that Fe had strong affinity to organic matter and could stabilize DOM by creating covalent bonds with specific molecular groups of humic molecules. Indeed, PVC-MPs treatment also exhibited higher dissolved Fe concentrations than RS. Another possible factor was microplastics stimulated the microbial activity in red soil and accelerated the decomposition of DOM. Here, however, these assumptions need further research. The paddy soils organic matter include microbial metabolites, animal and plant residues, semi-decomposed products of animal and plant residues (Chen et al., 2020), which are one of the most important Hg^{2+} complexing agents in soil. It can i) Inhibit Hg uptake by organisms by forming large complexes, ii) Act as a Hg "carrier" to facilitate dissolved or nanoparticle Hg transporting to membrane transport sites, iii) Serve as electron donor for Hg methylating bacteria (Bravo et al., 2017), or iv) Possibly act as a shuttle allowing Hg enter the cell when DOM components are used as carbon or energy source. In two paddy soils, the interaction between DOM and Hg may be an important reason for changing the bioavailable Hg^{2+} after PVC-MPs addition. SEM model showed that Comp 1 and Comp 3 were the closely related to bioavailable Hg^{2+}, which was similar to Taylor et al. (2019). Therefore, we concluded that the reduction of DOM fluorescent components intensities after adding microplastics was the main reason for the increasing content of bioavailable Hg^{2+} in red soil. There are also studies to proving organic matter had strong absorption to MeHg as well as Hg^{2+} and therefore acted to transport and sequester Hg (Siciliano et al., 2005), thereby limited Hg methylation.

We observed that PVC-MPs resulted in different trends in the amount of bioavailable Hg^{2+} in two soils (Fig. 2), whereas the bioavailable MeHg in both soils appeared to be inhibited in PVC-MPs treatments (Fig. 3). Wu et al. (2020b) reported that in agricultural (rice growing) wetlands, the spatiotemporal variation of Hg methylators activity had a larger effect on MeHg production variability than the variation in bioavailable Hg^{2+}. Members of methylators were significantly decreased with PVC-MPs addition in red soil, such as *Firmicutes* and *Proteobacteria*. In addition, the *Proteobacteria* identified in this research mainly included the *Gammaproteobacteria* and *Alphaproteobacteria* classes, which are nitrogen-fixing bacteria. These two classes were significantly decreased with microplastics amendment. Therefore, we concluded that the addition of microplastics affected soil net MeHg production by altering the soil microbial communities through changing the soil nutrient contents. We further confirmed this finding by identifying the genes associated with Hg methylation, but qPCR results were notably inconsistent with the methylation changes observed by 16S rRNA gene sequencing (Fig. 9), which had also been found in other recent studies (Ndu et al., 2018; Christensen et al., 2018). However, the addition of microplastics led to an increase in *hgc*A gene abundance, which may become a potential factor affecting the production of MeHg. Combined with SEM analysis results (Fig. 10), our research demonstrated that the increase of bioavailable Hg^{2+} in red soil after adding PVC-MPs was due to the decrease of fluorescent component related to Hg^{2+} and the change of reducing condition. The bioavailable Hg^{2+} did not increased with PVC concentration, and the possible speculation was that the concentration of PVC had reached the threshold for changing the bioavailable Hg^{2+}. The increase of DOC concentration and the decrease of sulfate reduction and methylation communities were the main factors leading to the decrease of bioavailable MeHg. In alkaline soil, dissolved Fe and the increase of fluorescent components related to Hg^{2+} were the main factors in soil causing the content of bioavailable Hg^{2+} and MeHg both decreased. In summary, differences in physiochemical and biological parameters of two paddy soils led to the distinction of Hg release after PVC addition.

It is worth noting that certain plastic-degrading species were discovered in this study. In fact, PVC-degrading species had already been reported (Pathak and Navneet, 2017). Similarly, reports of microbial biodegradable plastics in sediments have shown that *Isosphaeraceae* in sediments tends to strongly attach to the surface of polyurethane foams and inferred that it has the ability to degrade plastics (Fig. S7) (Jogler et al., 2020). We also observed this category in our red soils sample, and we speculated that it participated in microbial biodegradable plastics, based on the fact that its abundance in the microplastics treatment group was significantly higher than in RS group. Moreover, class *Anaerolineae*, *Ktedonobacteria* (Fig. S6) and family *Vicinamibacterales* (Fig. S7) were all higher in microplastics treatments than in RS. No researches to our knowledge, however, suggested hydrocarbon degradation capacity of these organisms. Therefore, further research is needed.

5. Conclusions and environmental implications

Amounts of plastics enter and are trapped in agricultural systems through plastic mulching, sewage sludge and compost. Here, we have demonstrated that PVC microplastics decreased the net MeHg contents in diverse paddy soils. In red soil, the increase of bioavailable Hg^{2+} after adding PVC-MPs was due to the decrease of fluorescent components related to Hg^{2+} and the change of reducing condition. The increase of DOC concentration and the decreases of sulfate reduction and methylation communities were the main factors leading to the decrease of bioavailable MeHg. In alkaline soil, dissolved Fe and the increase of fluorescent components related to Hg^{2+} were the main factors causing the contents of bioavailable Hg^{2+} and MeHg both decreased. Moreover, the leaching of chemicals from microplastics alone and itself have been shown to potentially contribute to the carbon pool in paddy soil and to influence microbial abundance and material cycle. Together, these results suggested that microplastics affected the migration and transformation of heavy metals by affecting biogeochemical cycles in soil, and improved our understandings on the role of microplastics in soil metal cycles. However, this study explored only a limited set of circumstances, namely, flooding paddy soil, a kind of plastic materials in relatively high doses. Further studies are needed considering the impacts of flooding and drought alternate, UV irradiation and types of microplastics impact on soil Hg methylation and demethylation, as well as taking into account the microbial degradation and photodegradation process of microplastics. Mercury is a serious and globally cycled contaminant, and its fate and ecological effects in paddy soils are threats to human health. More obviously, the effects of microplastics on Hg methylation should be taken into account when assessing the fate of Hg in paddy soils.

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CRediT authorship contribution statement

Xu Yang: Formal analysis, Investigation, Methodology, Software, Visualization, Writing – original draft, Writing – review & editing. **Zihao Li:** Formal analysis, Investigation, Methodology, Software. **Chi Ma:**

Investigation. **Zhongzhu Yang:** Investigation. **Jingjing Wei:** Investigation. **Tantan Wang:** Methodology. **Xin Wen:** Methodology. **Wenhao Chen:** Conceptualization, Data curation. **Xiaohong Shi:** Conceptualization, Supervision. **Yi Zhang:** Conceptualization, Funding acquisition, Project administration, Resources. **Chang Zhang:** Conceptualization, Funding acquisition, Project administration, Resources, Supervision.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supporting information

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jhazmat.2021.126895.

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